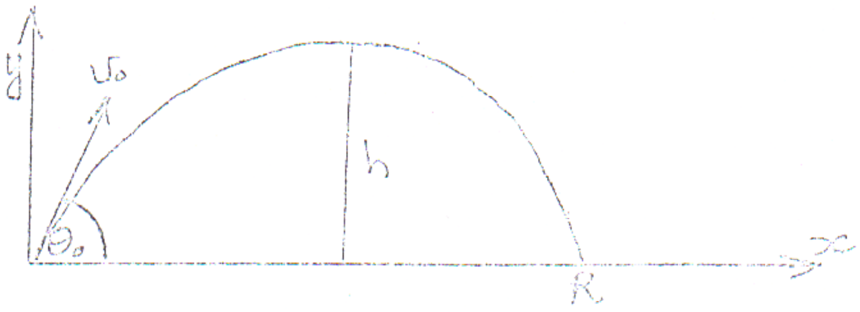
2003

VRAAG/*QUESTION 1*

(a) 'n Projektiel word oor 'n horisontale oppervlak gelanseer teen 'n hoek θ_0 met die horisontaal. Die beginsnelheid is v_0 en die maksimum trefafstand is R. Toon aan dat die vlugtyd gegee word deur $T = \frac{2\, v_0 \, \mathrm{Sin}\theta_0}{g}$.

A projectile is launched over a horisontal plane at an angle θ_0 relative to the horisontal. The initial velocity is v_0 and the range is R. Show that the time of flight is given by $T = \frac{2\, v_0 \, \mathrm{Sin}\theta_0}{g}$



By maks hoogte $(t = T/2)$ is $v_y = 0$

$\therefore v_y = (v_0 \sin\theta_0) - gt$

$0 = v_0 \sin\theta_0 - g(T/2)$

$\frac{g}{2} T = v_0 \sin\theta_0$

$T = \frac{2 v_0 \sin\theta_0}{g}$

(4)

(b) Toon aan dat die maksimum hoogte wat bereik word gegee word deur $h = \frac{v_0^2 \, \mathrm{Sin}^2\theta_0}{2g}$

Show that the maximum height is given by $h = \frac{v_0^2 \, \mathrm{Sin}^2\theta_0}{2g}$.

h is hoogte by $T/2$ $\quad y = (v_0 \sin\theta_0)t - \frac{1}{2} g t^2$

$h = v_0 \sin\theta_0 \left(\frac{2 v_0 \sin\theta_0}{2g}\right) - \frac{1}{2} g \left(\frac{2 v_0 \sin\theta_0}{2g}\right)^2$

$= \frac{v_0^2 \sin^2\theta_0}{g} - \frac{1}{2} \frac{v_0^2 \sin^2\theta_0}{g}$

$h = \frac{v_0^2 \sin^2\theta_0}{2g}$

(4)

Vraag / *Question* 1(c)(Vervolg)(*Continue*) / .....